

Nithish K Gnani

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Education

2013 - 2017 **Bachelor of Technology - Mechanical Engineering**, *National Institute of Technology Karnataka (NITK)*, Surathkal, Karnataka, India

Work Experience

2019 - Present **Research Assistant -> Technical Associate**, *Indian Institute of Science (IISc)*, Bangalore
Department of Electronic Systems Engineering & Center for Networked Intelligence

- Selected as a Technical Associate at Cisco - Center for Networked Intelligence (CNI), IISc. Recommended by a committee at CNI, under the division of EECS and Robert Bosch Center for Cyber-Physical Systems (RBCCPS)
- Leading industry-funded novel research projects which also includes planning, and procuring components while judiciously utilizing the available funds.

2017 - 2018 **Team Leader**, *Hindustan Coca-Cola Beverages Private Limited*, Hyderabad
Maintenance and Total Productive Maintenance (TPM)

- Managed a team of 20+ equipment associates to achieve consistent reliability of utility systems.
- Completed the graduate engineering trainee (GET) program with rotations in sales, road-to-market (shipping), and factory operations.

Projects

Mar' 2021 - Present **Time Sensitive Networking (TSN) and Tactile Cyber-Physical Systems**, *Indian Institute of Science*, Funders: Ministry of Electronics and Information Technology (MeitY), Govt. of India & Cisco CNI

- Objective: Design and implementation of a Tactile Cyber-Physical System (TCPS) for real-time interaction between humans and robots for applications requiring ultra-reliable low latency communication (uRLLC).
- Building an IEEE 802.1 TSN capable ethernet switch by implementing in hardware, a. time synchronization (IEEE 802.1AS), b. Time Aware Shaper (IEEE 802.1Qbv), and c. packet duplication and elimination (IEEE. 802.1CB).
- Novelty 1*: Developed P4TAS, a P4-based system implementation for offloading Time-Aware Shaper onto a programmable SmartNIC.
- Novelty 2*: Developed packet de-duplication algorithms for SmartNICs to efficiently eliminate duplicates for enhancing the reliability of Scheduled Traffic in Time-Sensitive Networks.
- Novelty 3*: Offloaded teleoperator intelligence to P4-programmable network edge switches to show that ports can be intelligent in ensuring physical actuation is satisfied in real-time and reducing network traffic by intelligently handling data packets with task-awareness.

Mar' 2020 - **AMMAZING**, *Indian Institute of Science*

- June 2022
- An end to end 5G mmWave system for infotainment.
 - Novelty*: Data & control plane of a video stream split over 5G and regular Wi-Fi.
 - Currently shortlisted in top 10 teams in ongoing 5G Hackathon by Dept. of Telecommunication, Government of India.

July 2020 - **Wi-Fi Modelling**, *Indian Institute of Science*, Funder: Boeing (ANRC)

- June 2021
- Modelling & simulation of IEEE 802.11n & 802.11ac using MATLAB.
 - Performance Metrics: throughput, packet error rate & range.
 - Implementing Rate Control Algorithms in MATLAB.

- Feb' 2020 - **Video Casting in dense environment**, *Indian Institute of Science*, Funder: Boeing
- Jan 2021
- Considering that 2.4 & 5 GHz Wi-Fi is congested in an aircraft cabin in which 400+ passengers use devices to cast media onto seat back displays in close proximity, the possibility of using IEEE 802.11ad, 60GHz Wi-Fi (Wi-Gig) & 802.11ax (Wi-Fi 6E) was explored.
 - WLAN Toolbox in MATLAB was used to simulate the physical behavior of Wi-Fi transmissions in a dense environment.
- June 2019 - **Airplane IoT data analytics and management**, *Indian Institute of Science*, Funder: Boeing
- Feb' 2020
- Developed machine learning algorithms to classify anomalies and take corrective actions using real-time data generated from sensors from a MATLAB/Simulink model of an aircraft environmental control system.
 - *Novelty*: Developed an algorithm that does linear interpolation between non-consecutive data points from time series data to adaptively store data based on a cost function which balances storage space savings and error in reconstruction of data.
- Mar' 2019 - **Acoustics Based Localization**, *Indian Institute of Science*, Under Dr. Chandramani Singh
- Jan' 2020 & Dr. T V Prabhakar, Funder: Boeing
- Developed an application to locate multiple wireless edge devices with embedded microphones by using audio signals from speakers.
 - Simulated a few localization algorithms using MATLAB. Implemented KNN fingerprinting based localization of the receivers and achieved an accuracy of 98%.
 - *Novelty*: Unlike the usual source localization, here, the receivers (mics) are localized when the locations of speakers are known.
- Nov' 2017 - **Total Productive Maintenance (TPM)**, *Hindustan Coca-Cola Beverages Pvt. Ltd.*
- Apr' 2018
- Implemented TPM steps 4 and 5 in the bottling plant which led to the **JIPM TPM Consistency Award**. This included the following work:
 - Conducting TPM audits, collecting data, and performing root cause analysis (RCA) for equipment failures
 - Training and coaching operators using one-point lesson (OPL) drives
 - Planning the cleaning, lubrication, inspection and tightening (CLIT) checklist and schedule
- Jul' 2016 - **Attitude Control of a CubeSat Using Reaction Wheels**, *National Institute of Technology, Karnataka*, Bachelor's thesis project
- Apr' 2017
- Designed a suspended body and achieved precise control of its yaw using reaction wheels using a manually tuned PID controller.
 - Modeled it in Simulink and implemented Model Reference Adaptive Control (MRAC) using the MIT rule and compared it with PID control.

Skills

Programming	P4, Python, MATLAB, C, Embedded C
Hardware	Netronome Agilio Smart NIC, Nordic nRF52 series, Arduino, NetFPGA-SUME
Others	Autodesk Fusion 360, 3D printing, Adobe Premiere Pro, Implementing papers
Languages	English (IELTS score 8.0), Hindi, Tamil, Kannada, Sourashtra

Publications

In-progress

1. **Nithish Krishnabharathi Gnani**, Joydeep Pal, Deepak Choudhary, Himanshu Verma, Soumya Kanta Rana, Kaushal Mhapsekar, T. V. Prabhakar, Chandramani Singh. EdgeP4: A P4-Programmable Edge Intelligent Ethernet Switch for Tactile Cyber-Physical Systems, (submitted to HotNets 2023)
2. Joydeep Pal, Deepak Choudhary, **Nithish Krishnabharathi Gnani**, Chandramani Singh, T. V. Prabhakar. P4TAS: A P4-based Time-Aware Shaper on SmartNICs, (submitted to 2023 IEEE Global Communications Conference (GLOBECOM))
3. Kaumudi Singh, **Nithish Krishnabharathi Gnani**, Pratyush Shukla, Sachin S M, T V Prabhakar, Joy Kuri, Judicious data management for sustaining an energy harvesting sensor node - extension (manuscript in preparation)

Published

1. Girish Vaidya, T.V.Prabhakar, **Nithish Gnani**, Ryan Shah, Shishir Nagaraja. A novel approach for identification of sensor devices through Acoustic PUF, *Digital Threats: Research and Practice*. <https://doi.org/10.1145/3488306>
2. Kaumudi Singh, Pratyush Shukla, Sachin S. M., **Nithish K. Gnani**, Prabhakar T. V., Joy Kuri. Judicious data management for sustaining an energy harvesting sensor node, *Concurrency and Computation: Practice and Experience*. <https://doi.org/10.1002/cpe.5997>

Certifications

1. Python Specialization – University of Michigan.
<https://coursera.org/verify/specialization/9LLM9T3Q55B8>
2. Deep Learning – DeepLearning.AI; Ongoing – 4/5 courses complete:
[https://coursera.org/verify/8PG2CRVLTEDH,45UFP3FNDV5G, BTQHB5N5CM7V,7899CQP7J2KS](https://coursera.org/verify/8PG2CRVLTEDH,45UFP3FNDV5G,BTQHB5N5CM7V,7899CQP7J2KS)

Volunteering

Taught middle and high school students at [Yuwa School](#) in Hutup Village in Jharkhand in February 2019. Coached students in tackling interviews. Prepared the team to attend the Laureus Awards 2019, where Yuwa won in "[Sports for Good](#)" category.

Interests

Swimming, PC gaming, 3D printing